THE UNIVERSITY OF MELBOURNE

ASSESSING

# QUALITY THINKING

NEW METRICS: 2025

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WITH MANY THANKS TO
schools, industry and credentialing research-practice partners, whose contributions continue to combine with the broader work of the University of Melbourne to inform the development of this work.

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# Assessment of Complex Competencies: A Poem

Assessment is
a **process**
of gathering evidence
from a **variety** of contexts
in which the learner can show what they are **capable** of
in a **complex** domain of learning
to support a **considered judgement**
about their position on a **scale** of competence
from less expert to more expert
representing what they know and can do
and what they need to learn next
with a **sufficient** degree of confidence
to support them in their learning
**and** to provide recognition of their attainment
that people can **trust**

- Enterprise Professor S. Milligan

THE UNIVERSITY OF MELBOURNE

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THE AMBITION

# TRANSFORMING WHAT WE VALUE IN SCHOOLS

**Current metrics in schooling favour narrow definitions of learning success, shaped by the industrial model, the old 'grammar of schooling', that are no longer sufficient in a modern education.**

—

**A new 'grammar of schooling' empowers young people to be producers of their own learning and supports them to develop the human competencies required to thrive in a world of great technological advancement, where they face global challenges and opportunities, and a range of unknown career options.**

Leading schools and school systems recognise that a broader range of metrics is required to support learners, teachers and school leaders to develop, assess and recognise the complex competencies young people need today.

The University of Melbourne has generated reliable and robust resources to support educational settings that have a strategic interest in:

- student agency in learning;
- advancing the way complex competencies are taught, assessed and reported on;
- credentialing learners at key transition points of schooling;
- engaging in professional learning to build staff capacity to lead new assessment practices;
- equipping students with sophisticated learning skills;
- contributing to the growing evidence-base for systemic change;
- using next-generation assessment and recognition approaches to leverage the transformation they wish to achieve.

Competency education is not an 'add on' to the existing features of schooling, rather an opportunity to learn knowledge and skills through developing and exercising competence in areas of interest.

Moving towards the new grammar of schooling is an opportunity to rethink the emphasis on standardisation in current practices and embrace the complexity of a more sophisticated student-centred approach to learning.

## OLD GRAMMAR → NEW GRAMMAR

Features of the old grammar are familiar to educators and help hold the industrial model of schooling in place (e.g., learning and success defined by age/stage, teacher solely as content expert, standardised examinations, timetabled blocks of learning).

Educators leading transformation consider the shifts required to move from the old grammar to the new. Below is a micro-level example of the kinds of shifts required in learning and teaching.

### OLD GRAMMAR FOCUS

Students **use feedback** to learn from mistakes and use teacher advice.

### NEW GRAMMAR FOCUS

Students **generate feedback loops** to seek, use and give feedback to enhance performance, learning skills, learning partnerships and autonomy.

THE AMBITION

# WHY COMPLEX COMPETENCIES?

**A shift to the new grammar of schooling with a commitment to the development of complex competencies makes it imperative that students develop agency in learning.**

Complex competencies might also be referred to as '21st Century skills'; 'soft skills'; 'general capabilities'; 'transferable skills', 'graduate attributes', 'approaches to learning', or similar. Despite the differences in name, all cover similar skills, attitudes and values that learners are expected to develop to thrive at school and in the future.

A complex competency is:

- teachable, learnable, observable, assessable;
- learned in the context of developing expertise in particular domains;
- developed inside and outside formal education;
- transferable across contexts, that is, competencies learned in one context or place that can be transferred to another*.

**Agency in Learning** is essential to the new grammar of schooling. This competency is considered fundamental to the development of all competencies.

**AGENCY IN LEARNING**

The capacity to produce learning of value to self or community

**ACTING ETHICALLY**

Acting in a way that is guided by moral principles (what we ought to do) in any situation

**ACTIVE CITIZENSHIP**

Contributing to the well-being of a community

**COMMUNICATION**

The ability to transmit, receive and interpret information, ideas, arguments, feelings and beliefs to support the purposes of the individual or group

**COLLABORATION**

Working with others to achieve a common goal

**QUALITY THINKING**

Thinking things through to achieve better outcomes for yourself or your group

**PERSONAL DEVELOPMENT**

The ability to live by the attitudes and behaviours that your communities value

FOR LEADERS

# CONDITIONS THAT SUPPORT A TRANSFORMATIVE AGENDA

**School leaders are in a position to optimise the conditions for teaching and learning of the complex competencies young people need in a modern education.**

Transforming the 'grammar of schooling' takes bold, nurturing and influential leadership. Experienced transformative leaders describe these actions as levers for change.

| LEADERS DO THIS... | SO THAT TEACHERS... | SO THAT LEARNERS CAN... |
|---|---|---|
| Champion learning ambitions that prepare learners for their life at school and beyond | Feel compelled and empowered to align their learning design and assessment practices | Feel empowered to learn according to their ambitions and interests |
| Privilege competencies as a focus of learning at the school | Design learning that includes a focus on learner competence | Have the opportunity to develop, exercise and demonstrate competence |
| Build a culture that values learner agency | Can be co-designers, coaches and facilitators of learning | Be empowered and responsible for their own learning and the learning of others |
| Consider the organisation of learning to prioritise relationships | Have the time and opportunity to work in depth with learners; to respond to their specific learning needs | Be truly known by their educators |
| Allow time and space for deep professional collaboration | Learn and work with colleagues from different ages, stages, learning areas, networks | Learn in transdisciplinary and multi-aged settings |
| Generate community interest and trust in the importance of competency education | Can work in partnership with students and families to develop learner competence | Be supported and recognised for what they know and can do, broadly defined, within and beyond the school |
| Prioritise next-gen assessment design | Can rethink the purpose of assessment | Know they are being recognised for a broad range of what they know and can do |

## A NOTE ON SCHOOL ASSESSMENT PRACTICES

It is important not to over-assess.

Complex competencies develop over time and with multiple exposures to opportunities to develop and exercise competence. They do not need to be measured at the beginning, middle and end of a unit of learning. It is recommended that teachers use the University's online assessment platform Ruby to assess a student in a particular competence no more than once or twice per year.

Unlike the attainment of domain-based knowledge or skills (e.g., in History or Maths), competence cannot be developed via direct instruction alone. Rather it is developed across a variety of occasions and contexts (e.g., collaboration can be developed in a Science class, a school musical, a group multi-domain inquiry project, etc).

School leaders align desired conditions for competence development with assessment protocols and schedules.

FOR LEADERS

# USING NEXT-GENERATION RECOGNITION

**Reporting and credentialing systems reflect what is valued in learning and can serve a variety of purposes for students, teachers and external stakeholders.**

## PURPOSE 1: PLANNING

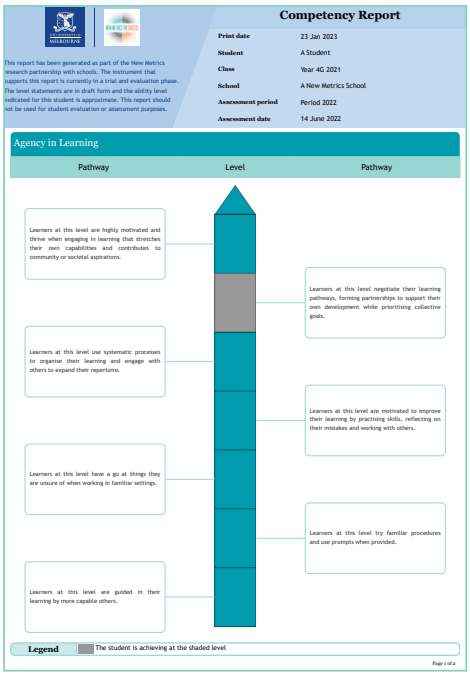
Teachers can use the University's online assessment platform, Ruby, to generate rocket reports for individual assessments of learners. These are useful for:

- planning 'next steps' for learners;
- informing learning design decisions;
- team discussions about learners and classes;
- reflection for teachers and learners.

Teacher planning also benefits from insights gained when learners share reflections on their own competence.

This serves the dual purpose of developing learner understanding of the competence (and why it is of value) and supporting teachers to plan for an individual learner's next steps.

A sample learner reflection tool is provided on p21. Before using such a tool, teachers should ensure learners understand the competence and its definition (through class discussions, sharing examples, etc) and how their reflections will be used to support their development.



This report has been generated as part of the New Metrics research partnership with schools. The instrument that supports this report is currently in a trial and evaluation phase. The level statements are in draft form and the ability level indicated for this student is approximate. This report should not be used for student evaluation or assessment purposes.

**Competency Report**

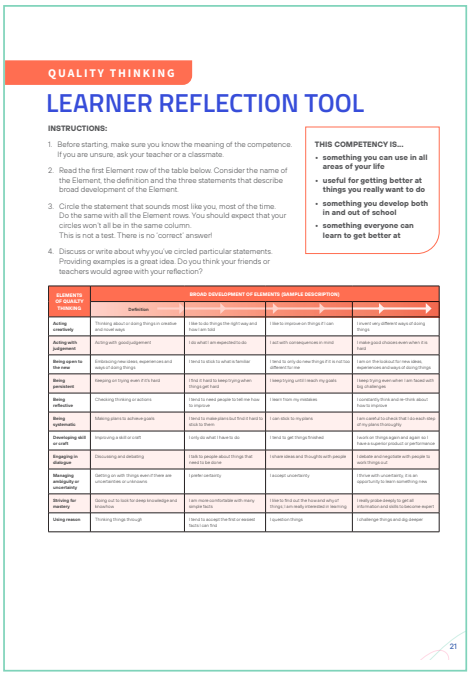
| | |
|---|---|
| Print date | 23 Jan 2023 |
| Student | A Student |
| Class | Year 4G 2021 |
| School | A New Metrics School |
| Assessment period | Period 2022 |
| Assessment date | 14 June 2022 |

**Agency in Learning**

| Pathway | Level | Pathway |
|---|---|---|
| Learners at this level are highly motivated and thrive when engaging in learning that stretches their own capabilities and contributes to community or societal aspirations. | | |
| | | Learners at this level negotiate their learning pathways, forming partnerships to support their own development while prioritising collective goals. |
| Learners at this level use systematic processes to organise their learning and engage with others to expand their repertoire. | | |
| | | Learners at this level are motivated to improve their learning by practising skills, reflecting on their mistakes and working with others. |
| Learners at this level have a go at things they are unsure of when working in familiar settings. | | |
| | | Learners at this level try familiar procedures and use prompts when provided. |
| Learners at this level are guided in their learning by more capable others. | | |

**Legend** The student is achieving at the shaded level

Page 1 of 2



QUALITY THINKING

**LEARNER REFLECTION TOOL**

**INSTRUCTIONS:**

1. Before starting, make sure you know the meaning of the competence. If you are unsure, ask your teacher or a classmate.
2. Read the first Element row of the table below. Consider the name of the Element, the definition and the three statements that describe broad development of the Element.
3. Circle the statement that sounds most like you, most of the time. Do the same with all the Element rows. You should expect that your circles won't all be in the same column. This is not a test. There is no 'correct' answer!
4. Discuss or write about why you've circled particular statements. Providing examples is a great idea. Do you think your friends or teachers would agree with your reflection?

**THIS COMPETENCY IS...**

- **something you can use in all areas of your life**
- **useful for getting better at things you really want to do**
- **something you develop both in and out of school**
- **something everyone can learn to get better at**

| ELEMENTS OF QUALITY THINKING | BROAD DEVELOPMENT OF ELEMENTS (SAMPLE DESCRIPTION) | | | |
|---|---|---|---|---|
| | Definition | | | |
| **Acting creatively** | Thinking about or doing things in creative and novel ways | I like to do things the right way and how I am told | I like to improve on things if I can | I invent very different ways of doing things |
| **Acting with judgement** | Acting with good judgement | I do what I am expected to do | I act with consequences in mind | I make good choices even when it is hard |
| **Being open to the new** | Embracing new ideas, experiences and ways of doing things | I tend to stick to what is familiar | I tend to only do new things if it is not too different for me | I am on the lookout for new ideas, experiences and ways of doing things |
| **Being persistent** | Keeping on trying even if it's hard | I find it hard to keep trying when things get hard | I keep trying until I reach my goals | I keep trying even when I am faced with big challenges |
| **Being reflective** | Checking thinking or actions | I tend to need people to tell me how to improve | I learn from my mistakes | I constantly think and re-think about how to improve |
| **Being systematic** | Making plans to achieve goals | I tend to make plans but find it hard to stick to them | I can stick to my plans | I am careful to check that I do each step of my plans thoroughly |
| **Developing skill or craft** | Improving a skill or craft | I only do what I have to do | I tend to get things finished | I work on things again and again so I have a superior product or performance |
| **Engaging in dialogue** | Discussing and debating | I talk to people about things that need to be done | I share ideas and thoughts with people | I debate and negotiate with people to work things out |
| **Managing ambiguity or uncertainty** | Getting on with things even if there are uncertainties or unknowns | I prefer certainty | I accept uncertainty | I thrive with uncertainty, it is an opportunity to learn something new |
| **Striving for mastery** | Going out to look for deep knowledge and knowhow | I am more comfortable with many simple facts | I like to find out the how and why of things I am really interested in learning | I really probe deeply to get at information and skills to become expert |
| **Using reason** | Thinking things through | I tend to accept the first or easiest facts I can find | I question things | I challenge things and dig deeper |

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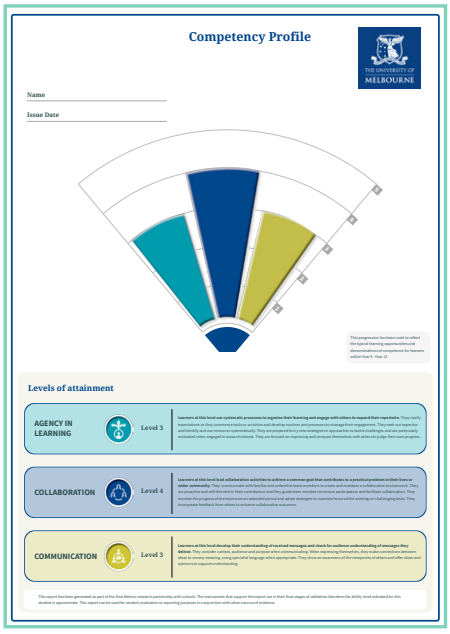
## PURPOSE 2: SCHOOL REPORTING

Fan Profiles represent aggregated assessments for each competency, for multiple competencies. These are useful for:

- yearly or semesterly reporting to learners and families;
- professional discussions about learners and learning design.

Learner fan profiles can be generated each semester if the following requirements are met:

- A minimum of two competencies must be assessed;
- For each competency, a minimum of two valid assessments must be completed.




## PURPOSE 3: CREDENTIALING

The Learner Competency Credential is issued to learners at key transition points in their education: primary to secondary; entry into senior secondary; end of senior secondary.

It captures aggregated assessments over time and recognises levels of learner attainment that are supported by robust moderation processes. It is formally warranted by The University of Melbourne.

Learners receive printable and digital credentials that have broad utility beyond their school for:

- transition from one educational setting to another (e.g., primary to secondary);
- admissions processes (e.g., tertiary entrance);
- internal qualifications (e.g., a Middle Years certificate);
- Employment.

Credentials can be generated at end of year if assessment and moderation requirements are met. Schools intending to generate credentials will work directly with the University team to understand these requirements and engage in a University-led school assessment audit.

**Image:** The Australian Learner Competency Credential

FOR LEADERS

# RUBY

**Ruby is the University's platform that schools use to conduct formal assessments and generate reports and credentials.**

### WHAT DOES RUBY DO?

Ruby invites teachers to check their judgements by answering a series of multiple-choice style questions about typical behavioural indicators of the competence being assessed. It uses teacher judgement to locate a learner on a standards-based learning progression and generates learner reports.

### HOW DOES RUBY WORK?

Ruby will prompt the user to nominate the purpose of the assessment i.e. teachers choose whether the assessment is for teacher planning, or for school reporting or credentialing.

Depending on the age / stage of learning, and the purpose of the assessment and reporting, Ruby will ask fewer or greater number of questions. For example, an assessment conducted for the purposes of teacher planning may have approx. 10 questions, whereas an assessment contributing towards a credential may have approx. 22 questions.

### WHAT REPORTS ARE GENERATED?

Regardless of the assessment purpose, all teacher assessments will auto-generate a rocket report. This includes teacher assessments that will contribute towards a learner fan profile or a credential.

## TEACHERS MAKE MICRO-JUDGEMENTS BY RESPONDING TO QUESTIONS ABOUT A LEARNER.

Questions in Ruby aim to bring to the surface indicative behaviours of competence across a range of ages, stages and contexts.

The example provided here shows a model of the language and the broad steps of progression teachers will select from when recording their judgements in Ruby.

**When constructing an argument, does the learner:**

- **state a position**
- **include a relevant range of points or ideas**
- **give the argument a logical structure**
- **build an argument that responds to possible objections**

FOR TEACHERS

# HOW TO SUPPORT LEARNERS TO DEVELOP COMPETENCE

### STEP 1.

**FAMILIARISE YOURSELF** with the competency. Use the following to strengthen a common understanding of the competency between teachers:

- competency definition (p13)
- age/stage appropriate learning progressions (p14 – 16)

In addition, it can be helpful to understand the composition of the competency

- elements of the competence (p17)
- broad development of elements (p18)

### STEP 2.

**CONSIDER LEARNING DESIGN** that provides opportunity for students to learn, exercise and demonstrate competence.

- Does it provide you with opportunity to observe evidence of learners' competence over time?
- Does it allow you to privilege learner agency in the development of competence?
- How can the competency be developed across multiple contexts in connection with other goals (e.g., developing competence whilst also attaining knowledge)?
- Do you have the opportunity to truly know each student as a learner?

Teachers can use the teaching for competence table on p12 to examine the conditions they provide to learners to exercise competence and activate their agency in learning.

Importantly, to demonstrate more advanced levels of competence (i.e., Levels 4 and 5 in the learning progression) students must be provided with conditions that activate their agency as learners and go beyond direct instruction or scaffolded teaching.

### STEP 3.

**ASSESS & REPORT.** Assessment must have a purpose. Determine the purpose of your assessment and have adequate evidence to support your judgement.

Refer to p8 & p9 to review reporting and credentialing options.

FOR TEACHERS

# TEACHING FOR COMPETENCE

Great teaching draws from a rich repertoire, using different approaches for a variety of purposes and learners. There are times when direct instruction is the most appropriate activity. There are also times teachers can be mentors that support learner agency through the conditions they provide. This table highlights features to be considered when providing students the opportunity to learn, exercise and demonstrate competence.

| Considerations for learning design and opportunity for agency | Purpose | Evidence of learning | Openness of process | Supervision | Scope of discipline knowledge and skill required | Authenticity | Scale | Connection to others |
|---|---|---|---|---|---|---|---|---|
| **TEACHER AS MENTOR** | Focuses on solving or participating in the solution of a practical problem for which the solution is unknown (i.e. goes beyond mastery of knowledge to application)<br><br>Contributes to something of interest in their lives or community. | Learner curates and designs evidence used to demonstrate skill (e.g., portfolio)<br><br>Evidence sourced inside and outside of school | The process is open and challenging, with little guidance provided by the teacher<br><br>Allows for learners to conceptualise, construct, organise, analyse and test their own process, and make their own mistakes | Learner mentored and advised on request | Learners need to be self-directed and collaborate to master new and deeper knowledge and skill | Replicates real world experiences, or is a real-world experience<br><br>Outcome has significance beyond the classroom | Requires long term (e.g., a semester) commitment and allows for flexible application<br><br>Allows revisiting, reflection and reconceptualization of ideas and mistakes with opportunity for immersion | Learners independently build connections with new people and places outside of school |
| **SCAFFOLDED TEACHING** | Shaped, but not defined, by a learner's interests<br><br>Includes choices within product or process | Learners negotiate how they will evidence their learning (e.g., product) | Process is defined with scope for interpretation and variation by the learner | Learner's work is supervised<br><br>Provided with guidance, examples, modelling, | Requires application and exploration of knowledge or skill in the related domain that goes beyond what the learner already knows and can do<br><br>Requires learner initiative to develop required knowledge or skill | Is relevant to the learners' immediate lives outside of the classroom | Scope for demonstrating depth and commitment (e.g., a five-week unit of work) | Learners independently connect with familiar people and places outside the classroom, (e.g., grandparents, neighbours) |
| **DIRECTED INSTRUCTION** | Goals, product, process, content determined by the teacher<br><br>Context selected by the teacher so it is likely to appeal to learners | Learners use provided templates to demonstrate learning (e.g., worksheet, template for a project) | Learner provided with familiar steps to follow | The learner is directed and supervised | Requires knowledge or skill that is mostly already known to the learner | Designed for classroom learning | Short, fixed duration (e.g., a focus for a few days) time limited, timetabled | Involves familiar school-based people<br><br>Connection with external others is planned and facilitate by the school |

# QUALITY THINKING

## Thinking things through to achieve better outcomes for the individual or group

QUALITY THINKING

# LEARNING PROGRESSIONS

**Assessors can recognise a learner's competence on a continuum of development known as a learning progression.**

The following learning progressions take into account a broad range of ages/stages of development. They represent the qualitative leaps in development of competence; reflecting the growth in sophistication of application of competence that arises as learners develop in age and throughout typical stages of schooling.

The progressions can be used to:

- develop teacher and learner understanding of growth in the competence;
- report on learner attainment and growth.

Competence is hierarchical and those at higher levels can and should still use the things they learned at lower levels to operate expertly in different contexts. Being at a higher level means one has at their disposal a wider and deeper repertoire to deploy as required.

The developmental levels are broad, as capacity to perform at a level derives from a complex amalgam of skills, knowledge, attitudes and values. No two individuals operating at the same level will act in the same way, as there are different ways to express their capabilities.

# KINDERGARTEN – YEAR 4

| Level | | Level Statement |
|---|---|---|
| 5 | THE INNOVATIVE THINKER | **Learners at this level can engage deeply with a topic, using good judgement and creativity to understand and explore ideas and to generate novel solutions.** They seek to understand situations in detail and are open to alternative conceptions and speculative explanations. They are able to support and test their conclusions with evidence and reason. They can create or innovate ideas by integrating diverse perspectives and can tolerate uncertainty and ambiguity. They are open to finding new conceptions to explain the 'how and why' of the world and exploring new ways to represent it. They are also open to critiques of their thinking and can refine or enhance their understanding through debate and feedback. |
| 4 | THE ANALYTICAL THINKER | **Learners at this level assemble arguments and evidence to explain, test, adapt or challenge ideas or representations.** They seek to identify relevant aspects of dilemmas and explore plausible solutions to open-ended problems. They employ sound judgement, generate ideas. and assess and counter different viewpoints. Probing for deeper understanding, they will question the validity of their own thinking in the light of evidence or critique. |
| 3 | THE INVESTIGATIVE THINKER | **Learners at this level are interested in investigating how and why things work, actively engaging in posing and solving problems.** They approach problems with openness and a commitment to trying a range of solutions, recognising that some problems may have more than one solution and can be approached in different ways. They investigate new concepts and experiment by drawing on connections from other problems or contexts. In responding to dilemmas or challenges to their thinking, they seek to justify their conclusions using arguments and some evidence regarding the pros and cons of their position. They recognise uncertainty and explore inconsistencies in their solutions by reflecting on their actions or approaches. |
| 2 | THE INQUISITIVE THINKER | **Learners at this level are curious about ideas and representations of the world, seeking understanding and solutions by asking questions.** They form opinions readily and adhere to familiar and proven thinking strategies. They can identify a range of elements in a problem situation and include some relevant points and ideas in their reasoning. They can describe the thinking behind their decisions and acknowledge the ideas of others. When their ideas or solutions are challenged, they provide reasons or examples which support their own view or approach. |
| 1 | THE STRUCTURED THINKER | **Learners at this level apply standard processes to solve familiar problems, accepting obvious conclusions or solutions as they are presented.** Their judgements are based on their own experience or past learning, and when expressing a thought or faced with uncertainty, they adhere to a stable position or opinion. In new situations, they identify obvious or superficial characteristics and use these to propose simple solutions to more complex issues. They respond to challenges to their ideas by describing or redoing procedural steps, seeking to justify their approach. |

QUALITY THINKING

# YEAR 5 – YEAR 8

| Level | | Level Statement |
|---|---|---|
| 5 | THE INNOVATIVE THINKER | **Learners at this level can engage deeply with a topic, using systematic processes, good judgement and creativity to understand and explore ideas, challenge orthodoxies, and generate novel solutions.** They seek to understand situations in detail and are willing to explore alternative conceptions and speculative explanations while supporting and testing their conclusions with evidence and reason. They can create or innovate ideas by integrating diverse perspectives and challenging assumptions. They accept uncertainty and ambiguity as they seek to find new conceptions to explain the 'how and why' of the world and explore new ways to represent it. They are sceptical of simple or received wisdom and open to critiques of their thinking. They can use debate and feedback to refine and enhance their understanding and approach. |
| 4 | THE ANALYTICAL THINKER | **Learners at this level assemble reasoned arguments and evidence to understand, test, adapt or challenge ideas or representations.** They seek to identify relevant aspects of dilemmas and explore plausible solutions to open-ended problems. They employ sound judgement and can generate ideas by synthesising concepts. They can assess and counter different viewpoints and construct cases addressing challenges to their views. Probing for deeper understanding, they will question the validity of their own thinking in the light of evidence or critique. |
| 3 | THE INVESTIGATIVE THINKER | **Learners at this level are interested in investigating how and why things work, actively engaging in posing and solving problems.** They approach problems with openness and a commitment to trying a range of solutions, recognising that some problems may have multiple solutions and can be approached in different ways. They investigate new concepts and experiment by drawing on connections from other problems or contexts. In responding to dilemmas or challenges to their thinking, they seek to justify their conclusions using arguments and evidence regarding the pros and cons of their position. They recognise uncertainty and explore inconsistencies in their solutions by reflecting on their actions or approach. |
| 2 | THE INQUISITIVE THINKER | **Learners at this level are curious about ideas and representations of the world, seeking understanding and solutions by asking questions.** They form opinions readily and adhere to familiar and proven thinking strategies. They can identify a range of elements in a problem situation and include a range of relevant points and ideas in their reasoning. They can describe the thinking behind their decisions and acknowledge the ideas of others. When their ideas or solutions are challenged, they can provide reasons, evidence, or examples which support their own view or approach. |
| 1 | THE STRUCTURED THINKER | **Learners at this level apply standard processes to solve familiar problems, accepting obvious conclusions or solutions as they are presented.** Their judgements are based on their own experience or past learning, and when arguing or faced with ambiguity, they adhere to a stable position or opinion. In new situations, they identify obvious or superficial characteristics and use these to propose simple solutions to more complex issues. They respond to challenges to their ideas by describing or redoing procedural steps, seeking to justify their approach. |

QUALITY THINKING

# YEAR 9 – YEAR 12

| Level | | Level Statement |
|---|---|---|
| 5 | THE INNOVATIVE THINKER | **Learners at this level can engage deeply with a topic, using systematic processes, good judgement and creativity to understand and explore ideas, challenge orthodoxies and generate novel solutions.** They analyse situations thoroughly, seeking out alternative conceptions and speculative explanations while supporting and testing their conclusions with evidence and reason. They create or innovate ideas by integrating diverse perspectives and challenging assumptions. They revel in uncertainty and ambiguity, seeking to find new conceptions to explain the 'how and why' of the world and explore new ways to represent it. They are sceptical of simple or received wisdom and open to critiques of their thinking. They use debate and feedback to refine and enhance their understanding and approach. |
| 4 | THE ANALYTICAL THINKER | **Learners at this level assemble reasoned arguments and evidence to understand, test, adapt and challenge ideas or representations.** They seek to identify all relevant aspects of dilemmas and explore plausible solutions to open-ended problems. They employ sound judgement and can generate ideas by synthesising concepts. They can assess and counter different viewpoints and construct cases addressing potential objections or challenges to their positions. Probing for deeper understanding, they will question the validity of their own thinking in the light of evidence or critique, and refine or adapt their ideas using alternative approaches or methodologies. |
| 3 | THE INVESTIGATIVE THINKER | **Learners at this level are interested in investigating how and why things work, actively engaging in posing and solving problems**. They approach problems with openness and a commitment to testing a range of solutions, recognising that some problems will have multiple solutions and may require shifts in perspective. They investigate new concepts thoroughly, and experiment by drawing on connections from other problems or contexts. In responding to dilemmas or challenges to their thinking, they justify their conclusions using arguments and evidence regarding the pros and cons of their position. They recognise uncertainty and explore inconsistencies in their solutions by reflecting on and evaluating their actions or approach. |
| 2 | THE INQUISITIVE THINKER | **Learners at this level are curious about ideas and representations of the world, seeking understanding and solutions by asking questions.** They form opinions readily and adhere to familiar and proven thinking strategies. They can identify a range of elements in a problem situation and include a variety of relevant points and ideas in their reasoning. They can articulate the thinking behind their decisions and acknowledge the ideas of others. When their ideas or solutions are challenged, they provide reasons, evidence, or examples which support their own view or approach. |
| 1 | THE STRUCTURED THINKER | **Learners at this level apply standard processes to solve familiar problems, accepting obvious conclusions or solutions as they are presented.** Their judgements are based on their own experience or past learning, and when arguing or faced with ambiguity, they adhere to a stable position or opinion. In new situations, they identify obvious or superficial characteristics and use these to propose simple solutions to more complex issues. They respond to challenges to their ideas by describing or redoing procedural steps, seeking to justify their approach. |

QUALITY THINKING

# ELEMENTS OF COMPETENCE

**The University of Melbourne's assessment architecture includes foundational elements of competence.**

Each complex competency is a combination of foundational elements of competence, wherein the whole (competency) is greater than the sum of its parts (elements).

When designing learning, teachers should primarily provide opportunity for the competence to be exercised and observed, rather than, for example, planning lessons that align with each element.

One might consider the elements as puzzle pieces or building blocks of the competence, or as vital organs that make a living thing.

Understanding these elements can be helpful in forming teachers conception of the competence, for inspiring learning design, and for tuning into the evidence of competence development.

The Elements (the 'building blocks' of competence)

The University's assessment architecture, including the learning progressions and elements, provides a 'currency' of competency assessment which can be used to assess a broad range of competencies.

This underpinning, standards-based assessment architecture allows students who are learning in different contexts, curricula and learning designs to be assessed and recognised in a robust and comparable way.

ELEMENTS OF

# QUALITY THINKING

**Thinking things through to achieve better outcomes for the individual or group**

DEFINITION:

### ACTING CREATIVELY

The ability to generate innovative or novel ways of thinking about or doing things

DEFINITION:

### ACTING WITH JUDGEMENT

The ability to act on the basis of sound assessments, evaluations or decisions

DEFINITION:

### BEING OPEN TO THE NEW

The ability to embrace or harness new ideas, experiences and ways of doing things

DEFINITION:

### BEING PERSISTENT

The ability to be tenacious in pursuit of a goal

DEFINITION:

### BEING REFLECTIVE

The ability to evaluate and learn from experience

DEFINITION:

### BEING SYSTEMATIC

The ability to act in accordance with a plan, method or system

DEFINITION:

### DEVELOPING SKILL OR CRAFT

The ability to hone, sharpen, polish or improve skill or craft

DEFINITION:

### ENGAGING IN DIALOGUE

The ability to engage in dialogue to connect to others, explore, negotiate meaning, or develop new perspectives

DEFINITION:

### MANAGING AMBIGUITY OR UNCERTAINTY

The ability to operate in the face of things that are unknown, uncertain, or undecidable

DEFINITION:

### STRIVING FOR MASTERY

The ability to develop deep expertise in a field or domain

DEFINITION:

### USING REASON

The ability to use thinking skills to explain, argue, analyse or evaluate

QUALITY THINKING

# UNDERSTANDING THE ELEMENTS OF COMPETENCE

It can be helpful to understand what broad development in the foundational elements of a competency looks like. This supports teachers to align their learning and assessment design.

It is important to remember that the synergy of the elements is what makes a complex competency. For example, whilst focusing on the development of a single element will indeed contribute to a learner's qualities, having a view to the marked leaps of development of the complex competence as a whole (the progressions) is crucial.

| ELEMENTS OF QUAILTY THINKING | BROAD DEVELOPMENT OF ELEMENTS (SAMPLE DESCRIPTION) | | | |
|---|---|---|---|---|
| | Definition | | | |
| **Acting creatively** | The ability to generate innovative or novel ways of thinking about or doing things | Uses the prevailing or standard approaches | Adds to, modifies, elaborates or improves the prevailing approaches | Generates or pioneers innovative, novel or imaginative approaches |
| **Acting with judgement** | The ability to act on the basis of sound assessments, evaluations or decisions | Acts on custom, rules, expectations or habit | Acts on the basis of consideration | Acts in a shrewd, practical, wise or discerning manner |
| **Being open to the new** | The ability to embrace or harness new ideas, experiences and ways of doing things | Acknowledges new things (e.g., ideas, views, approaches) | Engages with or tries out new things (e.g., ideas, views, approaches) | Seeks out new things (e.g., ideas, views, approaches) |
| **Being persistent** | The ability to be tenacious in pursuit of a goal | Makes an effort | Stays on task | Is unrelenting in commitment to a task despite challenges and setbacks |
| **Being reflective** | The ability to evaluate and learn from experience | Thinks about actions or experiences | Evaluates thinking or action when outcomes do not match expectations | Takes lessons from all experiences and reflects to improve |
| **Being systematic** | The ability to act in accordance with a plan, method or system | Uses intuitive or ad-hoc strategies | Follows a set procedure | Methodically works through options to chart an optimal path |
| **Developing skill or craft** | The ability to hone, sharpen, polish or improve skill or craft | Works or produces as asked or directed | Works or produces to reach a finishing point or pre-defined standard | Works or produces continuously, experimenting to extend or improve craft, skill, method or technique |
| **Engaging in dialogue** | The ability to engage in dialogue to connect to others, explore, negotiate meaning, or develop new perspectives | Conducts transactional interactions | Connects with others to share ideas, information or feelings | Negotiates understandings, interprets ideas, and explores points of view to develop novel or shared perspectives |
| **Managing ambiguity or uncertainty** | The ability to operate in the face of things that are unknown, uncertain, or undecidable | Acknowledges uncertainty | Accepts uncertainty or undecidability | Treats uncertainty or undecidability as opportunity to explore alternative possibilities |
| **Striving for mastery** | The ability to develop deep expertise in a field or domain | Focuses on gathering facts and surface skills | Inquisitive and investigative about the 'why' or 'how' of phenomena in a field or domain | Develops a deep and nuanced level of expertise in a field or domain |
| **Using reason** | The ability to use thinking skills to explain, argue, analyse or evaluate | Bases thinking on specific examples or personal experience | Applies general principles or abstract concepts | Displays scepticism: questions assumptions, identifies biases, evaluates evidence, critiques arguments |

QUALITY THINKING

# LEARNER REFLECTION TOOL

**INSTRUCTIONS:**

1. Before starting, make sure you know the meaning of the competence. If you are unsure, ask your teacher or a classmate.
2. Read the first Element row of the table below. Consider the name of the Element, the definition and the three statements that describe broad development of the Element.
3. Circle the statement that sounds most like you, most of the time. Do the same with all the Element rows. You should expect that your circles won't all be in the same column. This is not a test. There is no 'correct' answer!
4. Discuss or write about why you've circled particular statements. Providing examples is a great idea. Do you think your friends or teachers would agree with your reflection?

**THIS COMPETENCY IS...**

- **something you can use in all areas of your life**
- **useful for getting better at things you really want to do**
- **something you develop both in and out of school**
- **something everyone can learn to get better at**

| ELEMENTS OF QUAILTY THINKING | BROAD DEVELOPMENT OF ELEMENTS (SAMPLE DESCRIPTION) | | | |
|---|---|---|---|---|
| | Definition | | | |
| **Acting creatively** | Thinking about or doing things in creative and novel ways | I like to do things the right way and how I am told | I like to improve on things if I can | I invent very different ways of doing things |
| **Acting with judgement** | Acting with good judgement | I do what I am expected to do | I act with consequences in mind | I make good choices even when it is hard |
| **Being open to the new** | Embracing new ideas, experiences and ways of doing things | I tend to stick to what is familiar | I tend to only do new things if it is not too different for me | I am on the lookout for new ideas, experiences and ways of doing things |
| **Being persistent** | Keeping on trying even if it's hard | I find it hard to keep trying when things get hard | I keep trying until I reach my goals | I keep trying even when I am faced with big challenges |
| **Being reflective** | Checking thinking or actions | I tend to need people to tell me how to improve | I learn from my mistakes | I constantly think and re-think about how to improve |
| **Being systematic** | Making plans to achieve goals | I tend to make plans but find it hard to stick to them | I can stick to my plans | I am careful to check that I do each step of my plans thoroughly |
| **Developing skill or craft** | Improving a skill or craft | I only do what I have to do | I tend to get things finished | I work on things again and again so I have a superior product or performance |
| **Engaging in dialogue** | Discussing and debating | I talk to people about things that need to be done | I share ideas and thoughts with people | I debate and negotiate with people to work things out |
| **Managing ambiguity or uncertainty** | Getting on with things even if there are uncertainties or unknowns | I prefer certainty | I accept uncertainty | I thrive with uncertainty, it is an opportunity to learn something new |
| **Striving for mastery** | Going out to look for deep knowledge and knowhow | I am more comfortable with many simple facts | I like to find out the how and why of things; I am really interested in learning | I really probe deeply to get all information and skills to become expert |
| **Using reason** | Thinking things through | I tend to accept the first or easiest facts I can find | I question things | I challenge things and dig deeper |

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